

Sol1:

```
#include <iostream>
using namespace std;

class SerialObject {
private:
    static int count;
    int serialNumber;

public:
    SerialObject() {
        count++;
        serialNumber = count;
    }

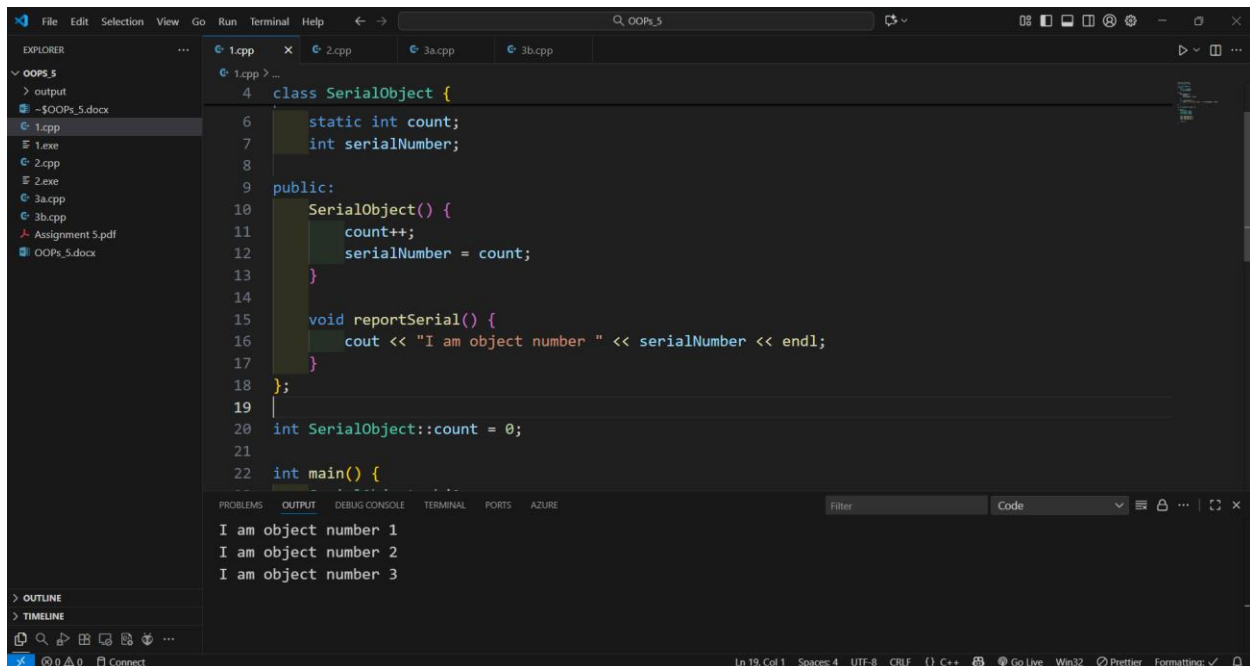
    void reportSerial() {
        cout << "I am object number " << serialNumber << endl;
    }
};

int SerialObject::count = 0;

int main() {
    SerialObject obj1;
    SerialObject obj2;
    SerialObject obj3;

    obj1.reportSerial();
    obj2.reportSerial();
    obj3.reportSerial();

    return 0;
}
```



Sol 2:

```
#include <iostream>
#include <iomanip>
using namespace std;

class Time {
private:
    int hours;
    int minutes;
    int seconds;

public:
    Time() : hours(0), minutes(0), seconds(0) {}

    Time(int h, int m, int s) : hours(h), minutes(m), seconds(s) {}

    void display() const {
        cout << setfill('0') << setw(2) << hours << ":"
              << setw(2) << minutes << ":"
              << setw(2) << seconds << endl;
    }

    void addTime(const Time& t1, const Time& t2) {
        seconds = t1.seconds + t2.seconds;
        minutes = t1.minutes + t2.minutes + seconds / 60;
        seconds %= 60;
        hours = t1.hours + t2.hours + minutes / 60;
        minutes %= 60;
    }
};
```

```

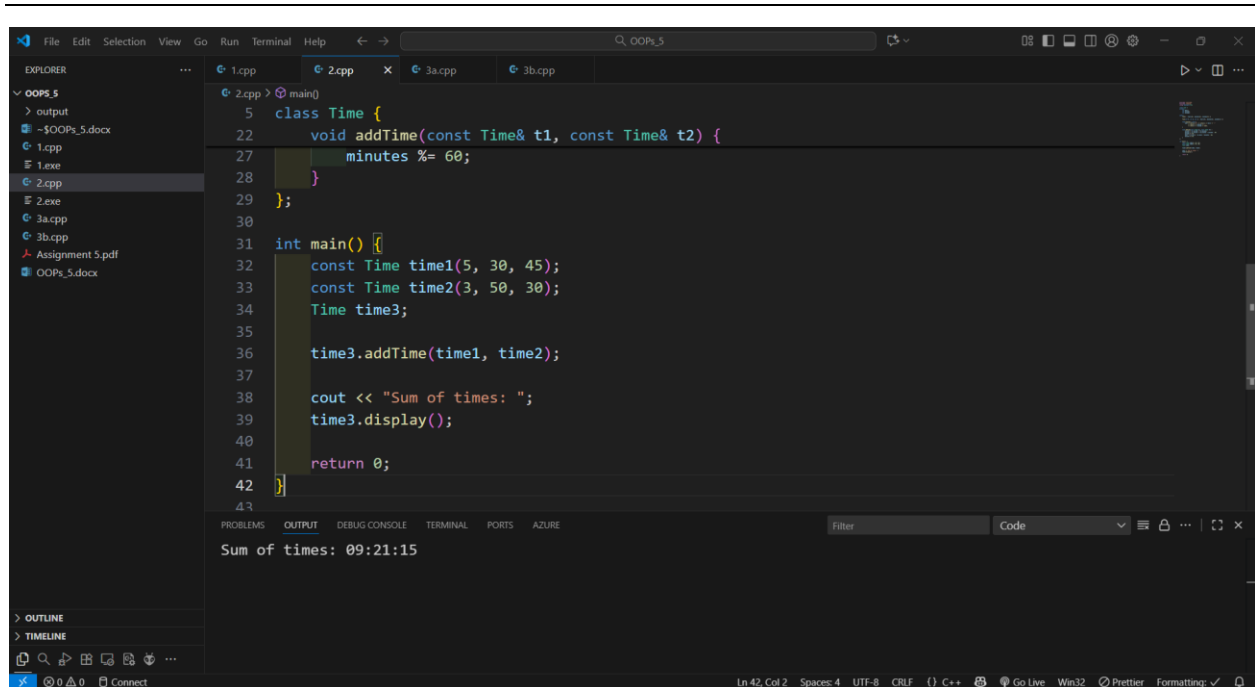
int main() {
    const Time time1(5, 30, 45);
    const Time time2(3, 50, 30);
    Time time3;

    time3.addTime(time1, time2);

    cout << "Sum of times: ";
    time3.display();

    return 0;
}

```



Sol 3a:

```

#include <iostream>
using namespace std;

int main() {
    int studentClass, failedSubjects, graceMarks = 0;

    cout << "Enter the class obtained (1/2/3): ";
    cin >> studentClass;

    cout << "Enter number of subjects failed: ";
    cin >> failedSubjects;

    switch(studentClass) {
        case 1:
            if(failedSubjects <= 3)
                graceMarks = 5 * failedSubjects;
            else

```

```

        graceMarks = 0;
    break;

case 2:
    if(failedSubjects <= 2)
        graceMarks = 4 * failedSubjects;
    else
        graceMarks = 0;
    break;

case 3:
    if(failedSubjects <= 1)
        graceMarks = 5 * failedSubjects;
    else
        graceMarks = 0;
    break;

default:
    cout << "Invalid class entered" << endl;
    return 1;
}

if(graceMarks > 0)
    cout << "Grace marks awarded: " << graceMarks << endl;
else
    cout << "No grace marks awarded" << endl;

return 0;
}

```

The screenshot shows the Visual Studio Code interface with a C++ file named 3a.cpp open. The code in the editor is as follows:

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int studentClass, failedSubjects, graceMarks = 0;
6
7      cout << "Enter the class obtained (1/2/3): ";
8      cin >> studentClass;
9
10     cout << "Enter number of subjects failed: ";
11     cin >> failedSubjects;
12
13     switch(studentClass) {
14         case 1:
15             if(failedSubjects <= 3)
16                 graceMarks = 5 * failedSubjects;
17             else
18                 graceMarks = 0;

```

The OUTPUT window at the bottom shows the execution results:

```

d:\Github\DataStructuresAndAlgorithms\NGIPT\OOPS\OOPS_5\output>.\"3a.exe"
Enter the class obtained (1/2/3): 1
Enter number of subjects failed: 1
Grace marks awarded: 5

```

The Explorer panel on the left shows the project structure for OOPS_5, including files like 1.cpp, 2.cpp, 3a.cpp, 3b.cpp, 1.exe, 2.exe, 3a.exe, and 3b.exe, along with a folder named output.

Sol 3b:

```
#include <iostream>
#include <iomanip>
using namespace std;

class Angle {
private:
    int degrees;
    float minutes;
    char direction;

public:
    Angle() : degrees(0), minutes(0.0), direction('N') {}

    Angle(int deg, float min, char dir) : degrees(deg), minutes(min),
direction(dir) {}

    void getAngle() {
        cout << "Enter degrees: ";
        cin >> degrees;
        cout << "Enter minutes: ";
        cin >> minutes;
        cout << "Enter direction (N/S/E/W): ";
        cin >> direction;
    }

    void displayAngle() const {
        cout << degrees << (char)248 << fixed << setprecision(1)
        << minutes << "' '" << direction << endl;
    }
};

int main() {
    Angle angle1(149, 34.8, 'W');

    cout << "Initial angle: ";
    angle1.displayAngle();

    char choice;
    do {
        Angle userAngle;
        cout << "\nEnter angle details:" << endl;
        userAngle.getAngle();

        cout << "You entered: ";
        userAngle.displayAngle();

        cout << "\nEnter another angle? (y/n): ";
        cin >> choice;
    } while(choice == 'y' || choice == 'Y');

    return 0;
}
```

